



Examination Report

Mathematics

6880

EGCSE

2025

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General comments

This paper was prepared for core candidates only, covering assessment objectives in the core syllabus only.

Overall, candidate performance in this examination declined noticeably compared to the previous year. In 2025, a striking number of candidates—101 in total—scored zero, a sharp increase from just nine in the previous year. This reflects a significant drop in overall achievement. In addition, no candidate managed to score 60 or above, and many obtained marks in the single digits. Only a small number of candidates, fewer than ten, achieved scores above 50, which further emphasizes the generally low performance observed.

Questions which proved to be easier for the candidates were: 1(a)(i), 2(c), 7(a), 11(a), 13 and 18(a): and those questions which proved to be difficult to the candidates were: 3(c), 5(a)(i), (ii), 5(b), 7(b), 8(b) 10, 11(b).

A number of candidates failed to show necessary working which resulted to loss of method marks. Some candidates used a calculator even in questions where it was specifically stated that a calculator should not be used. Several learners still struggled with spellings, e.g. naming polygons. Another trait observed was the use of equal sign to compare two unequal sides or omitting the equal sign where it has to be used.

Another shortfall observed was the use of inspection to solve equations resulting in further loss of marks.

Also candidates failed to present their work correctly, what they wrote meant a different thing which resulted to a loss of marks.

For an example

$$\frac{3 + 11 + 19 + 23 + y + 67}{6}$$

$$= 27$$

The above is an expression which cannot be solved but

$$\frac{3 + 11 + 19 + 23 + y + 67}{6} = 27$$

Is an equation which can be solved.

The above shows that candidate knew the method to use but failed to present it correctly.

Comments on specific questions

Question 1

(a) Candidates were given the fraction $\frac{132}{64}$ to write as a

(i) mixed number

Correct answer: $2\frac{1}{16}$

Generally well answered

Common wrong answers: 2.063, 2.0625

- Confusing decimals with mixed numbers

(ii) percentage

Correct answer: 206.25

Common wrong answers: 206.3, 206

(b) Candidates were ordered to work out 3.157^2 correct to 2 decimal places.

Correct answer: 9.97

Fairly done.

- Some candidates multiplied 3.157 by 2

Common wrong answers: 9.10, 9.96, 3.16, 6.316, 10.

Question 2

(a) Candidates were required to find the highest common factor of 72 and 48.

Correct answer: 24

Fairly done.

- Some candidates gave LCM as an answer.

Common wrong answers: any common factor, 144, 2, 6, 8, 12.

(b) Candidates were asked to simplify the ratios:

(i) 72 : 48

Correct answer: 3 : 2

Well done.

- Some candidates did not divide by HCF.

Common wrong answer: 6:4, 2:3, 1.5, 9:6

(ii) 7 m: 42 cm

Correct answer: 50:3

Poorly done.

- Learners divided without converting to common units first.
- Some included units in their final answer
- Others couldn't leave ratio in simplest terms

Common wrong answers: 1:6, 50cm:3cm,

(c) Candidates were given the ratio of length and width of a rectangle, length: width=3:2.

They were required to find the length of the rectangle if the width is 8 cm.

Correct answer: 12 cm

It was well done

Common wrong answers; 24, 13, 5.3 .

Question 3

Candidates were given the vectors: $\underline{a} = \begin{pmatrix} 4 \\ -8 \end{pmatrix}$, $2\underline{b} = \begin{pmatrix} -6 \\ 12 \end{pmatrix}$ and $\underline{a} - \underline{c} = \begin{pmatrix} 3 \\ -10 \end{pmatrix}$ and they were asked to find:

(a) $\underline{b}$

Correct answer: $\begin{pmatrix} -3 \\ 6 \end{pmatrix}$

Fairly done.

Common wrong answers: $\begin{pmatrix} -12 \\ 24 \end{pmatrix}$, $\begin{pmatrix} 3 \\ 6 \end{pmatrix} - \frac{3}{6}$

(b) $\underline{c}$

Correct answer: $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Fairly done

- Candidates failed to add directed numbers.

Common wrong answers: $\begin{pmatrix} 1 \\ -18 \end{pmatrix}$, $\begin{pmatrix} 7 \\ -18 \end{pmatrix}$, $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$

(c) the magnitude of a .

Candidates were expected to show their working and give their answer correct to three significant figure or better.

$$\sqrt{4^2 + (-8)^2} = \sqrt{80}$$

Correct answer: 8.94

It was poorly done

- Candidates did not put brackets
- They did not use square-root sign until at the end.

Common wrong answers: 8.9

Question 4

Candidates were asked to calculate the duration, n , in months for a loan given amount borrowed =E500, total paid=850 and simple interest rate 10% per month.

They were expected to divide the total interest by the interest per month.

Correct answer: 7

Fairly done.

- Some candidates divided 850 by 50

Common wrong answers: 17, 6, 7 years

Question 5

(a) (i) Candidates were asked to write 216 as a product of its prime factors in index form.

Correct answer: $2^3 \times 3^3$

Poorly done.

Common wrong answers: $1 \times 2^3 \times 3^3$, $2^3 3^3$, $2 \times 2 \times 2 \times 3 \times 3 \times 3$

(ii) Candidates were required to evaluate $16^{\frac{3}{2}}$ without using a calculator.

They were expected to use the laws of fractional indices, i.e. rewriting given numbers using root sign then evaluate as shown below.

$$16^{\frac{3}{2}} = (\sqrt{16})^3 = 4^3$$

Correct answer: 64

Poorly done.

- They used calculators
- Others thought its a mixed number.

Common wrong answer: 24, 35, 17.5,

(b) Candidates were expected to apply the law of zero index to evaluate $\left(\frac{-14x^2y}{7xy}\right)^0$.

Correct answer: 1

Poorly done.

- Some candidates simplified the fraction only.

Common wrong answers: $2x$, $-2x$, 0 .

Question 6

Candidates were given some rows of a Pascal triangle and required to write down the next row.

Correct answer: 1 7 21 35 35 21 7 1

Fairly done

- Some candidates used 35 as symmetry
- Some failed to use **next** as following but gave the above row.
- While others used diagonals to get the sequence.

Common wrong answer: 1 3 3 1 ; 1 7 26 34 31 15 1 ; 1 7 21 35 21 7 1

Question 7

(a) Candidates were asked to write the mathematical name for a polygon with 8 sides.

Correct answer: Octagon

Well done

- Some candidates had problems with spelling.

Common wrong answers: heptagon, regular octagon, oxtagone, octagonal.

(b) Candidates were asked to find the number of sides of a regular polygon with an interior angle = 140° . They were expected to use the sum of exterior angles of a polygon as shown:

$$\frac{360}{180 - 140}$$

Correct answer: 9 sides

Poorly done.

- Most candidate were calculating sum of interior angles

Common wrong answers: 2 sides, names of polygons, 1290

(c) Candidates were required to construct an equilateral triangle with side 4cm.

They were expected to use a compass on their construction without erasing their construction arcs.

Correct answer: Equilateral triangle of side 4cm with visible construction arcs.

Fairly done.

- Some candidates used a protector and measured 60°
- While others faked arcs

Common wrong answers: drew any quadrilateral, isosceles triangle, a line

Question 8

A rectangle ABCD with side CD= 6cm and diagonals AC and BD meeting at O was given. It was also given that angle AOB = 48° .

(a) Candidates were asked to find

(i) Angle CBO

Correct answer: 24°

Poorly done.

Common wrong answers: 48, 66, 28, 132.

(ii) Angle OCD

Correct answer: 66°

Fairly done.

Common wrong answers: 24° , 132° , 48°

(b) Candidates were given the area of triangle AOD as 12 cm^2 and asked to find the length of side AD.

They were expected to use the formula of the area of a triangle, substitute known values and solve for unknown as shown below

$$\frac{1}{2} \times AD \times 3 = 12$$

$$3AD = 24$$

Correct answer: 8

Poorly done.

- Some candidates used formula for area of a rectangle
- Others used 6 *cm* as height.

Common wrong answers: 12, 4.

Question 9

Candidates were supposed to identify quadrilaterals with given symmetry properties.

They were asked to write down the name a quadrilateral that has

(a) order of rotational symmetry 2 and no line of symmetry.

Correct answer: Parallelogram

Poorly done.

Common wrong answers: kite, trapezium, rhombus, any shape they remembered.

(b) Order of rotational symmetry 2 and 2 lines of symmetry.

Correct answer: Rectangle or Rhombus

Poorly done.

Common wrong answers: square, kite, rectangular.

Question 10

Candidates were asked to find duration of a rainfall that started falling at 1925 hours and stopped at 0423 the following day.

They were expected to subtract starting time from finishing time, with 24 hours added to finishing time since it is the next day.

Correct answer: 8 hours 58 minutes

Poorly done.

- Some candidates subtracted 1925 – 0423
- While others had difficulty subtracting time which overlapped to the next day.

Common wrong answers: 1502, 8hrs 48min, 9hr 58min

Question 11

Candidates were given the function, $f(x) = 2x - 1$.

They were asked to find

(a) $f(2)$

Candidates were expected to substitute 7 for x in the function and work out.

Correct answer: 13

It was well done.

Common wrong answer: $f(13)$, 3, 14 -1, $13x$, $x = 13$

(b) $f^{-1}(x)$

Candidates were to find the inverse using algebra method (or flow chart), showing their working clearly.

$$y = 2x - 1$$

$$y + 1 = 2x$$

$$x = \frac{y+1}{2}$$

Correct answer: $\frac{x+1}{2}$

Poorly done

- some candidates did not understand the $f^{-1}(x)$, $f(-1)$

Common wrong answers: -3 , $\frac{y+1}{2}$, $\frac{x}{2} + 1$, $x + \frac{1}{2}$

Question 12

Candidates were given a grid with a straight line through (1,0) and (0,-1)

(a) Candidates were asked to find the equation of the give line.

Correct answer: $y = x - 1$

It was fairly done.

- Some candidates calculated y - intercept yet they were expected to read it.

Common wrong answers: $y = -1$, $x = 1$, $y = x$, $x = -1$, $y = -x - 1$

(b) Candidates were asked to state the gradient of the line with equation $y = 3x - 5$.

Correct answer: 3

Poorly done

Common wrong answers: $3x$, -5 , others drew a line

Question 13

Candidates were given a simple linear equation, $2x + 5 = 29$ to solve. They were expected to show all their working as shown below.

$$2x + 5 = 29$$

$$2x = 24$$

Correct answer: 12

Well done.

- Some candidates worked out $29 - 5 = 34$
- Others missed the equal sign, hence a loss of marks,
- While others solved by inspection.
- Others divided 29 by 5.

Common wrong answers: 5.8, $\frac{29-5}{2}$, $x = 17$,

Question 14

Candidates were given an expression to expand brackets and simplify.

$$3(2x + 5) - 4(x - 2)$$

$$6x + 15 - 4x + 8$$

Correct answer: $2x + 23$

Fairly done

- Some candidate failed to evaluate -4×-2 , they gave 8.
- Some multiplied the first term only by 3.

Common wrong answers: $10x + 23$, $2x + 7$, $2x + 13$.

Question 15

Candidates were given simultaneous equations to solve.

The most convenient method was using the substitution method but any clear method was acceptable.

$$y = 3x$$

$$y = x - 8$$

Solution:

$$3x = x - 8$$

$$2x = -8$$

Correct answers: $x = -4$ and $y = -12$

Poorly done.

- Most candidates loved to use elimination method yet they were not good at it.
- Some substituted for $y = 0$.

Common wrong answers: $x = 4$ and $y = 12$; $x = 3$ and $y = -8$

Question 16

Candidates were given the expression, $2px - 6py$ to factorise.

They were expected to factorise the expression fully.

Correct answer: $2p(x - 3y)$

Fairly done.

- Most candidate did partial factorisation hence they got 1 mark.

Common wrong answers: $2(xp - 3py)$, $p(2x - 6y)$,

Question 17

Candidates were given a distribution and asked to complete an ordered stem-and-leaf plot to show the distribution.

They were expected to write the leaf part as the key and stems were given.

Correct answer:

Stem	Leaf					
0	4	6	7	9		
1	3	6	7	7	8	9
2	0	2	3	6	7	
3	1	3	3	4	8	9
4						

Poorly done

- Most candidates did not show knowledge of stem and leaf as they left blank spaces,
- Others would draw a leaf and a stem,
- While others use commas to separate each leaf,
- Others did not order the stem.

Common wrong answers: tally marks, leaves, frequency.

Question 18

Candidates were given a list of marks scored by a group of learners in a test. The marks were arranged in order of size starting with the smallest first as show

x 11 19 23 y 67

(a) Candidates were given that the range is 64 and they were required to find the value of x .

Correct answer: 3

Well done.

Common wrong answers: 64, 5, 56, -3

(b) Candidates were given that the mean is 27. They were asked to calculate the value of y .

They were expected to show all necessary working to find the value of y

Working:

$$\frac{3 + 11 + 19 + 23 + y + 67}{6} = 27$$

$$\frac{123 + y}{6} = 27$$

$$123 + y = 162$$

Correct answer : 39

Poorly done.

- Some candidates knew they had to add and divide by 6 but they could not add properly, they got $123y$ instead of $123 + y$.
- Some candidate presented their work wrongly (not acceptable).

Common wrong answers: 20.5, 6.5, 14, 3.5 (median position)

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Key Message

Candidates must show workings for all answers, especially answers to questions that are worth more than one mark. They must also give their answers to the required level of accuracy.

General Comments

This paper was prepared for core candidates only, covering assessment objectives in the core syllabus only. Surprisingly, a reasonable number of candidates scored above 70. There was also a reasonable number of candidates who scored marks in the thirties and twenties. There were also a few candidates scoring zeros after attempting most of the questions, some writing a few wrong answers, and leaving most answer spaces blank.

Generally all questions were attempted by candidates and were accessible to all since there were no questions that were left unattended by candidates. Time allocated for this component of the paper proved to be enough since most candidates were able to finish.

This paper provided a wide range of syllabus objectives. These objectives allowed candidates to prove how well they understood the concepts taught. Most questions required the candidates to produce what was learnt in class, other than applying it. There were no 'show' or 'prove' questions. This proved to be an advantage to most candidates who understood the concepts very well.

The presentation of solutions showed a decline yet again in this cohort. There was an increase to the number of candidates who gave correct answers without showing any working, and this led to the candidates losing both the method marks and the accuracy marks. There was also a number of candidates that showed premature approximations, which led to inaccurate answers, causing them to lose the accuracy marks.

There was also a new problem of candidates resorting to solving problems by finding missing information in an equation by trial, other than using the algebraic methods that are covered in the syllabus. e.g. they were required to solve $4x - 3 = -31$, quite a number of them gave $4(-7) - 3 = -31$, hence giving answer as $x = -7$.

Teachers are advised to encourage candidates to show all necessary workings clearly, as well as use a suitable level of accuracy. They should also encourage them to use their calculators fully during this paper, with proper interpretations.

Questions that were easily accessible to most candidates were **Question 2(d), Question 3(a), Question 5(a), Question 9(a), Question 11(a), and Question 14(a)**. Questions that proved to be the most challenging to candidates were **Question 1(a)(iii), Question 4(c), Question 5(b), Question 7(a) and (c), Question 8(a), Question 9(b)(i)(ii), Question 10 (a)(ii), Question 12(a)(ii), and Question 13(a) and (d)**.

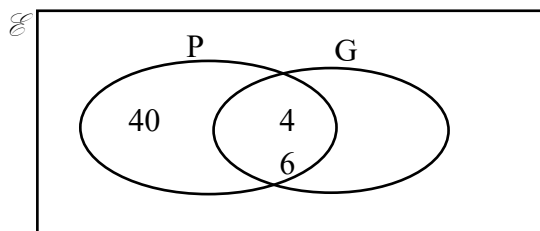
Comments on Specific Questions**Question 1**

- (a) *A form 5 class has 104 students.
60 students take Geography.
86 students take Physics.*

The Venn diagram shows part of the information.

$P = \{\text{Physics students}\}$

$G = \{\text{Geography students}\}$



(i) Find the number of students who take Geography only.

This part of the question was well answered by most candidates.

Common wrong response: 18, (from $104 - 40 - 46$).

Expected response: 14

(ii) Find the number of students who take neither Geography nor Physics.

This part was poorly done by most candidates. Some were not able to interpret the question and the diagram correctly.

Common wrong responses: 46 and 0

Expected response: 4

(iii) Find $n(P \cap G)'$.

This part was poorly answered by most candidates. They were not able to interpret the set symbols correctly. Some candidates ignored the complement symbol, and some ignored the n which means the number of elements in the set and decided to list the numbers in the regions.

Common wrong responses: 46, {4, 14, 40}

Expected response: 58

(b) $\xi = \{1, 2, 3, 4, \dots, 20\}$

$A = \{\text{Prime numbers}\}$

$B = \{\text{factors of } 60\}$

List members of

(i) A,

This part was well answered. Most candidates still had the misconception of 1 being prime and 2 not being prime. Some candidates did not include 19 in their sets. This led to loss of marks. Some were listing odd numbers, or even numbers.

Common wrong responses: { 1,3,5,7,11,13,17,19}, {1,3,5,7,9,11,13,15,17,19}

Expected response: {2, 3, 5, 7, 11, 13, 17, 19}

(ii) $A \cap B$,

This part was poorly done due to the fact that their set A was not correct, because of the exclusion of 2 or the inclusion of 1, and this led to the loss of the mark. Most candidates though, were able to list the correct elements of set B .

Common wrong response: $\{1, 3, 5\}$, $\{1, 2, 3, 5\}$, $\{3, 5\}$

Expected response: $\{2, 3, 5\}$

(iii) $A \cap B'$.

This part was also poorly done since it also dependent on their response from members of set A . A few candidates had a challenge in interpreting the compliment symbol.

There were no common wrong responses since different responses were seen.

Expected response: $\{7, 11, 13, 17, 19\}$.

Question 2

Write 6753

(a) *to the nearest 100,*

This part of the question was well done by most candidates. A few candidates did not maintain the place values in their answers, leading to loss of marks.

Common wrong responses: 68, 6700, 7000, 6750

Expected response: 6800

(b) *to 1 significant figure,*

This part was fairly done by most candidates. A few candidates wrote a single digit to represent the 1 significant figure.

Common wrong responses: 7, 700, 6000

Expected response: 7000

(c) *in standard form,*

This part was poorly done. Some candidates had no idea what was expected of them. The few that remembered the concept had either the wrong power of ten or the wrong value of a , forgetting that a must be in the range $1 \leq a < 10$. Some were even writing an expanded form of the number. All these led to loss of marks.

Common wrong responses: 6.753×10^{-3} , 6.8×10^3 , 0.6753×10^k , 67.53×10^k ,
6000 + 700 + 50 + 3.

Expected response: 6.753×10^3

(d) in words.

This part was well done by most candidates, if we ignore the spelling. They had a hard time writing the correct spelling. The most challenge was in writing thousand, hundred, fifty and three. Some candidates decided to write the number in part **(b)** or part **(c)** in words, and they lost marks.

Common wrong spellings: fifty, fifty, firthy, hundret

Expected response: Six thousand seven hundred and fifty three.

Question 3

(a) Find 27% of E8600.

This part was well done by most candidates. They were able to calculate the percentage of a quantity. Some had a challenge in interpreting the number from their calculators, ending up writing it as a decimal. Others decided to add the original amount to the answer.

Common wrong responses: 2'322 (from calculator reading), 232 200 (from multiplying by 27 instead of 27%), 10 922 (from adding 2 322 to 8600).

Common wrong workings: $\frac{27}{8600} \times 100$, $\frac{8600}{27} \times 100$, $\frac{127}{100} \times 8600$

Expected response: $\frac{27}{100} \times 8600 = 2\,322$

(b) Lombuso invests E3700 in a savings account.

The account pays 7% annual compound interest.

Calculate the total amount in the account at the end of 5 years.

This question was fairly done by most candidates. Most of them used the long method, calculating the investment for each year, up to 5 years, which led to an inaccurate answer due to premature rounding off. Some candidates had the correct formula for compound interest but had a challenge in applying the power 5, instead they multiplied by the 5. A few candidates calculated simple interest.

Common wrong workings: $\left(1 + \frac{7\%}{100}\right)^5 \times 3700$, $\left(\frac{1+7}{100}\right)^5 \times 3700$

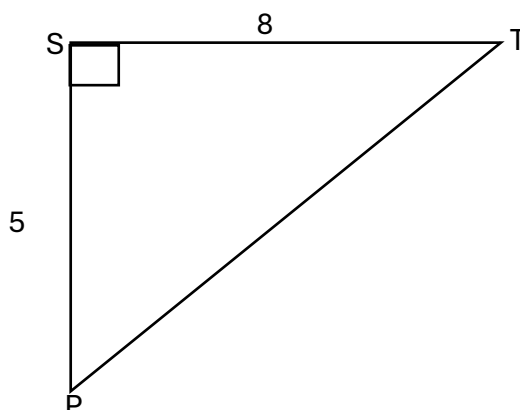
Common wrong responses: 4995 (from simple interest), 19 795 (from $1.07 \times 3700 \times 5$)

Common inaccurate responses that did not earn marks: 5180, 5189.40, 5189.45

Expected response: $3700 \times \left(1 + \frac{7}{100}\right)^5 = 5189.44$

Question 4

PST is a right angled triangle.



(a) *Calculate the length of PT.*

This question was fairly well answered by most candidates. Most of them were able to use the Pythagoras theorem correctly. Their challenge was giving the answer to 3 significant figures as expected, and they lost the accuracy mark. A few struggle to apply it correctly.

Common wrong working: $8^2 + 5^2 = \sqrt{8^2} + \sqrt{5^2} = 8 + 5 = 13$, $\sqrt{8^2 - 5^2} = \sqrt{39} = 6.24$

Common wrong responses: 9.4, 6.24, 13

Expected response: $\sqrt{8^2 + 5^2} = \sqrt{89} = 9.43$

(b) *Calculate angle STP.*

This part was poorly done. Most candidates had an idea that they need to use trigonometry but failed to apply it correctly. Some even tried to use the sine rule or the cosine rule. Most of them used the trigonometric ratios but they were mixing up the sides. A few used their calculators in the gradient mode leading to incorrect responses. Some decided to use the sine ratio or the cosine ratio which required them to use their hypotenuse from part **(a)**, which was not even correct.

Common wrong workings: $\tan 90^\circ = \frac{8}{5}$, $\sin T = \frac{5}{8}$, $\frac{180-90}{2} = 45$

Common wrong answers: 57.99 (from using $\tan^{-1} \frac{8}{5}$), 35.6 (from using correct ratio but with calculator in gradient mode)

Expected response: $\tan^{-1} \left(\frac{5}{8} \right) = 32.0^\circ$

(c) *The triangle PST is the cross-section of a prism.*

The length of the prism is 25 cm.

Calculate the volume of this prism.

This part of the question was poorly done by most candidates. Most candidates did not use the correct formula for finding the volume of a prism, instead, they mostly used the formula for volume of a cuboid, some even for a pyramid. Some that had the correct formula were not able to find the cross sectional area. Some even attempted to find the surface area of the prism.

Common wrong workings: $5 \times 8 \times 25$, $\frac{1}{3} \times 8 \times 5 \times 25$, $\frac{1}{2} \times (8 + 5) \times 25$

Common wrong responses: 1000, 333.3, 162.5

Expected response: $\frac{1}{2} \times 8 \times 5 \times 25 = 500$

Question 5

Candidates were given a grid where both axis were labelled from -8 to 10. On the grid there was triangle A at (3,1), (3,5), (1,3), triangle D at (-3,-2), (-3,-6), (-4,-4) and triangle E at (1,-3), (3,-1), (5,-3)

(a) *Reflect triangle A in the y-axis. Label the image B.*

This part was well done. Most candidates were able to reflect the object in the given mirror line correctly. A few reflected in the line $x = -1$. There were no common errors.

Expected response: Triangle with vertices at (-1,3), (-3,1), (-3,5).

(b) *Draw the image of triangle A after an enlargement, scale factor 3, centre (0, 3).
Label the image C.*

This part was poorly done by most candidates. They just could not perform enlargement. The few that can enlarge either enlarged using the wrong center or the wrong scale factor. Some did not even answer the question.

Expected response: A triangle at (3,3), (9,-3), (9,9).

(c) *Describe fully the **single** transformation that maps
(i) triangle A onto triangle D,*

This part was fairly done by candidates. The most challenge was the spelling of 'translation', it came in every form e.g translation, translocation was the most common, transaction. Other candidates were still stuck with sliding. Giving the translation vector was also a challenge, some gave it as a coordinate, which lost marks. Some described the vector in words e.g 6 steps to the left and 7 steps down, which did earn some marks though.

Common wrong vectors: (-6,-7), $\begin{pmatrix} -7 \\ -6 \end{pmatrix}$, $\begin{pmatrix} 6 \\ 7 \end{pmatrix}$, $\begin{pmatrix} -6 \\ -3 \end{pmatrix}$.

Expected response: Translation, $\begin{pmatrix} -6 \\ -7 \end{pmatrix}$.

(ii) *triangle A onto triangle E.*

This part was fairly done. Candidates were able to identify the transformation as rotation, but failed to give the center. Some gave the coordinates of the center without brackets e.g. 0,0 and they lost marks for that. Some even failed to give the direction of rotation after identifying the rotation of 90° . They were struggling between clockwise and anticlockwise using the signs. Some gave -90° anticlockwise.

Expected response: Rotation, (0,0), 90° clockwise (or -90°).

Question 6

The positions of port A and port C were shown in a diagram for candidates.

A ship leaves port A on a bearing of 065° and sails for 16 km to port B.

The ship then sails from port B to port C.

(a) *Use a scale of 1 cm for 2 km to construct a scale drawing connecting the ports A, B and C.*

This part of the question was fairly well done by most candidates. They were able to convert the 16 km to cm. They were also able to measure the bearing of 065° and to measure the length of 8 cm. The challenge was the accuracy of the measurements.

Expected response: an angle of 65° , measured clockwise from North of A and the line AB to be 8cm.

(b) *Find the bearing and distance of port C from port A.*

This part was also fairly well done by most candidates. The challenge was the accuracy of the measurements. Some read the wrong scale of the protractor, instead of reading 120° , they read 60° . A few candidates read the bearing from the anticlockwise direction. The other challenge was the units of the distance, some decided to give the distance in centimeters instead of kilometers.

Common wrong responses for the bearing: 240° , 115° , 300° , 295°

Expected response: distance = 13.8 km, bearing = 120°

(c) *Find the distance of port C from port B.*

This part was well done as a follow through from their diagram. Most candidates did not score marks because they gave answer in centimeters instead of kilometers.

Expected response: 14 km

Question 7

Musa's home is 900 m from the bus station.

Musa walked from home, a distance of 600 m steadily for 5 minutes.

He stopped for 1 minute to search for his bus ticket.

He then ran with a constant speed until he reached the bus station in 1 minute.

Candidates were given a distance-time graph. The distance was given in meters, from 0 to 900. The time was given in minutes, from 0 to 9 minutes. The first part of the journey, walking 600m for 5 minutes, was drawn on the graph.

(a) Complete the next 2 stages of Musa's journey in the diagram above.

This part was poorly done by most candidates. The 2nd stage of the journey was well done, they were able to draw a horizontal line from (5,600) to (6,600). A few drew longer lines up to (7, 600). The last part of the journey was poorly done. Most of the candidates joined the last part of their 2nd journey to (9,900), which was the last coordinate on the graph. Others decided to return back home, with last coordinate being (7, 0) or (9, 0).

Expected response: A line from (5,600) to (6,600), another line from (6,600) to (7,900)

(b) Find the distance travelled in the first 3 minutes.

This was well done by most candidates. They were able to read the corresponding distance at 3 minutes. Some candidates had a challenge in reading the scale properly, which led to incorrect answers. There were a few candidates that decided to find the area under the curve.

Common wrong responses: 306, 350, 330, 300.6, 380, 540 (from area under curve.)

Expected response: 360

(c) Calculate Musa's speed in the last part of the journey.

Write your answer in m/s.

This part was poorly done by most candidates. The challenge was their wrong graphs from part (a). Most candidates did not even have an idea that they were supposed to calculate the gradient of the journey. The few that had an idea of the gradient had a challenge with the units, since the graph was in minutes yet they were supposed to find the speed in m/s.

Common wrong responses: 1.667, 10, 300

Expected response: 5 m/s

Question 8

(a) Factorise $18 - 2y^2$.

This question was poorly done. Most candidates were able to identify 2 as a common factor. They then failed to identify the difference of two squares. Most of them did partial factorisation other than factorising fully. Candidates were able to earn some marks for the correct partial factorisation .

Common wrong responses: $2(9 - y^2)$, $9 - y^2$, $2(y^2 - 9)$, $2y(9 - y)$, $2(9 - y)(9 + y)$

Expected response: $2(3 - y)(3 + y)$

(b) Solve.

(i) $\frac{t}{4} - 1 = 5$

This part of the question was fairly well answered. Most candidates were able to solve the fractional linear equation. Some candidates had a challenge in clearing the fraction, resulting in wrong solutions.

Common wrong workings: $4t - 4 = 20$, $t - 1 = 20$, $4t - 1 = 5$, $\frac{t}{4} - \frac{1}{4} = \frac{5}{4}$

Common wrong responses: 6, 21, 1.5, 16

Expected response: $t - 4 = 20 \rightarrow t = 24$ or $\frac{t}{4} = 6 \rightarrow t = 24$

(ii) $x^2 - 2x = 32 + 2x$

This part was poorly done by most candidates. They failed to write the equation in the form $ax^2 + bx + c = 0$ first, but were seen factorizing to double brackets their equation which was still in the form $ax^2 + bx = c$. They were able to find the necessary factors of 32, which they used to factorise. Some coincidentally factorized correctly and found the correct answers, and were able to earn marks. Some attempted the quadratic formula or to complete the square which resulted to wrong solution in most cases.

Common wrong working: $x^2 - 4x = 32 \rightarrow (x + 8)(x - 4) = 32 \rightarrow x + 8 = 32$, $x - 4 = 32$
 $\rightarrow (x + 8)(x - 4) = 0 \rightarrow x + 8 = 0$, $x - 4 = 0$

Common wrong responses: - 8 and 4, 24 and 36, 40 and 28

Expected response: $x^2 - 4x - 32 = 0 \rightarrow (x + 4)(x - 8) = 0 \rightarrow x = 8$ and $x = -4$.

(iii) $2x - 7 < 13$

This part was well done by most candidates. A few candidates changed the sign to an equal sign or to greater than, hence lost some marks.

Common wrong responses: $x = 10$, $x > 10$, $x < 3$, $x \leq 10$

Expected response: $2x < 20 \rightarrow x < 10$

Question 9 This concept on functions was poorly done by most candidates. Most candidates were just fumbling through the question. They did not know when to detach the name of the function from the expression of the function.

$f(x) = 4x - 3$ and $g(x) = x + 6$.

(a) (i) Evaluate $f(4)$.

This part was fairly done by most candidates. They were able to substitute 4 correctly into $f(x)$. There were no common errors.

Expected response: $(4 \times 4) - 3 = 13$

(ii) $f(a) = -31$. Find the value of a .

This part was poorly done. Most candidates substituted the - 31 as an input, yet it was an output. The ones that were able to identify it as an output solved it by trial instead of using algebra.

Common wrong working: $(4 \times -31) - 3 = -127$

Common wrong responses: -127 , 8.5 (from $f(a) = 31$).

Expected response: $4a - 3 = -31 \rightarrow a = -7$

(b) Simplify.

(i) $f(x) - g(x)$

This was poorly done by most candidates. Candidates substituted the correct function but did not put brackets around $g(x)$, which was supposed to result in the change of sign. Some candidates also had a challenge in the addition of directed numbers.

Common wrong working: $f(4x - 3) - g(x + 6) \rightarrow 4x - 3 - x + 6$

Common wrong responses: $3x + 3$, $3x - 3$, $3x + 9$

Expected response: $4x - 3 - (x + 6) \rightarrow 4x - 3 - x - 6 \rightarrow 3x - 9$

(ii) $\frac{g(x)}{5} - \frac{f(x)}{2}$

This part of the question was poorly done. Most candidates could not simplify the expression to a single fraction. Most of them multiplied the expression with the lowest common multiple, leaving the final answer with no denominator. They also had a challenge in opening brackets with a negative number. Some candidates failed to substitute the correct functions, instead they swapped them.

Common wrong workings:

$$\frac{g(x+6)}{5} - \frac{f(4x-3)}{2} \rightarrow 2(x+6) - 5(4x-3) \rightarrow 2x + 12 - 20x - 15 = -18x - 3$$

Common wrong responses: $\frac{-18x-3}{10}$, $\frac{3x-36}{10}$, $-18x + 27$, $-18x + 3$

Expected response: $\frac{(x+6)}{5} - \frac{(4x-3)}{2} \rightarrow \frac{2x+12-20x+15}{10} \rightarrow \frac{-18x+27}{10}$

Question 10

(a) (i) Expand and simplify. $(x-3)(2x+2)$

This part was fairly done by most candidates. They proved to know the process of opening double brackets. Their challenge was opening brackets with a negative number. Others had a challenge of multiplying x by x . Others opened the brackets correctly then had a challenge in collecting like term, especially the term with x and x^2 .

Common wrong workings: $2x^2 + 2x - 6x + 6$, $2x^2 + 2x - 6x + 1$, $2x + 2x - 4x - 6$

Common wrong responses: $2x^2 - 4x + 6$, $2x^2 - 4x + 1$, $4x^2 - 6x + 6$, $x^2 - 2x - 3$

Expected response: $2x^2 - 4x - 6$

(ii) Simplify $(2x^2)^3$

This part was poorly done, the correct response was rarely seen. Most candidates were adding the indices. Some candidates just multiplied the 2 and the index 2 then raised that to power 3.

Common wrong responses: 64, $64x$, $8x$, $8x^2$, 8, $2x^6$, $2x^5$, $6x^2$

Expected response: $8x^6$

(b) $h = 12f + 3$

(i) Rearrange the formula to make f the subject.

This question was fairly done by most candidates. They were able to make f the subject. Some left their answer with a negative denominator, $\frac{3-h}{-12}$, and the lost some marks.

Common wrong responses: $\frac{-3h}{12}$, $\frac{3+h}{12}$, $\frac{3-h}{12}$, $f = 12h + 3$

Expected response: $\frac{h-3}{12}$

(ii) Find the value of f when $h = 27$.

This part was well answered by most candidates, even if they got part (i) wrong. They used the original formula to find the value of f . It indicated that candidates did not have a link in solving algebraic equations and numerical equations. They literally did the same steps they needed to do in part (i), to find the value of f . Other candidates with the correct expression for f in part (i) did not use their expression to find f but rather used the original formula.

Common wrong response: -2 , 327 (from $12h + 3$), 2.5 (from $\frac{27+3}{-12}$), $26\frac{3}{4}$ (from wrong use of the calculator when evaluating $\frac{27-3}{12}$)

Expected response: $\frac{27-3}{12} = \frac{24}{12} = 2$

Question 11

(a) Write down the order of the matrix $\begin{pmatrix} -1 & 3 \\ 2 & 0 \\ 7 & 4 \\ 0 & 4 \end{pmatrix}$.

This question was well done by most candidates, they were able to give the order of the matrix correctly.

Common wrong responses: 2 by 4, (4,2), 4 rows, 2 columns

Expected response: 4 by 2

$$(b) \ 2 \begin{pmatrix} 2 & 3 \\ 4 & 5 \\ -1 & -1 \end{pmatrix} - \begin{pmatrix} 4 & 1 \\ -2 & 6 \\ y & y \end{pmatrix} = \begin{pmatrix} 0 & x \\ 10 & 4 \\ -5 & z \end{pmatrix}$$

Find the values of x , y and z .

This part was fairly done. Candidates were able to find the values of the variable, some even without showing detailed working. Most of the candidates solve by trial instead of finding equation and solving algebraically. A common error was ignoring the 2 which was a scalar for the first matrix, which led to wrong solutions. Some candidates found two different values of y since y appeared twice in the matrices.

Common wrong responses: $x = 2$ $y = -1, -9$ $z = 1, 7$

Expected response: $6 - 1 = x \rightarrow x = 5$ $-4 - y = -5 \rightarrow y = 1$ $-2 - y = z \rightarrow z = -3$

Question 12

The ages of 90 people who visited a game park on a particular day were recorded.

The table summarizes the information.

Age	Number of people
$10 \leq x < 15$	8
$15 \leq x < 20$	18
$20 \leq x < 25$	22
$25 \leq x < 30$	15
$30 \leq x < 35$	11
$35 \leq x < 40$	
$40 \leq x < 45$	12

(a) (i) *Complete the table.*

This part was well done by most candidates. A common error was assuming that there were a 100 people.

Common wrong responses: 14, 10

Expected response: 4

(ii) *Find the percentage of the total number of people who were younger than 25.*

This question was fairly done. Candidates were able to find the percentage correctly. The challenge was finding the number of people who were younger than 25 years, some found 26, and some found 22, instead of 48.

Common wrong workings: $\frac{26}{90} \times 100$, $\frac{22}{90} \times 100$, $\frac{48}{100} \times 90$

Common wrong responses: 28.9%, 24.4%, 43.2%, 48%

Expected response: $\frac{48}{90} \times 100 = 53.3\%$

(b) Part of a histogram is drawn on the grid below to represent the information.

Candidates were given a histogram on a grid with the frequency density numbered up to 28 and the age from 10 to 45. The first 3 bars were drawn for them.

Complete the histogram.

This part was well done by the candidates. Their challenge was reading the vertical scale correctly, especially for the odd numbers. They read their 15 at 15.2 and their 11 at 11.2. Candidates were expected to add the bars for $25 \leq x < 30$ up to $40 \leq x < 45$. Some candidates then missed the bar for $25 \leq x < 30$, and started from $30 \leq x < 35$, but on the same space for $25 \leq x < 30$, thus shifting the positions for all the remaining bars, and they lost all the marks.

Expected response: A bar for $25 \leq x < 30$ with a height of 15, a bar for $30 \leq x < 35$ with a height of 11, A bar for $35 \leq x < 40$ with a height of 4, and a bar for $40 \leq x < 45$ with a height of 12.

Question 13

A class took a Mathematics test and a Biology test.

Marks for the learners are shown in the table.

Mathematics	90	80	71	56	44	42	32	21	10
Biology	87	84	64	50	49	45	38	18	12

Candidate were given a scatter diagram, with Biology marks on the vertical axis and Mathematics marks on the horizontal axis. The first 7 data points were plotted on the scatter diagram.

(a) *The first 7 data points are plotted in the scatter diagram.*

Plot the last two data points.

This part was poorly done due to the challenge candidates had in interpreting scale, especially the odd data points. Most of them were able to plot (10,12). They had a challenge with plotting (21,18), most of them plotted at (22,18).

Expected response: Plotted coordinates at (21,18) and a plotted coordinate at (10,12).

(b) *State the type of correlation shown by the scatter diagram.*

This part was not well done, most candidates seemed not to have an idea what was expected of them. There were no common wrong responses.

Expected response: Positive correlation.

(c) *Draw a line of best fit for the diagram.*

This part was fairly done. Candidates were able to draw a line through the points. Some drew a line with all points below or all points above. Some had most of the points on the same side. Some candidates just connected the points using free hand or a ruler. All these lines did not earn any marks.

Expected response: A straight line through the points such that there is an equal number of points on either side of the line.

(d) *Mbuso got 49% in Mathematics but did not take the Biology test.
Estimate the mark Mbuso would have obtained in Biology.*

This part was poorly done by candidates. Most of them did not have the line of best fit. The few that had the line of best fit did not use it to find the corresponding mark for Biology. A common wrong answer was 44, which corresponds to the 49 for Biology from the given data.

Common wrong working: $\frac{49}{90} \times 100 = 44.1\%$

Common wrong responses: 44, 44.1, 42, 52

Expected response: 45 to 49

14 *A group of 480 people were asked which hot beverage they liked.
The results are summarized in the Pie Chart.*

Candidates were given a pie, which was NOT to scale, chart illustrating the sectors of the four beverages. The sector angles were not given.

(a) *There were 100 people who liked coffee.
Find the sector angle for people who liked coffee.*

This part was fairly well done. Candidates were able to find the sector angle, using the correct method.

Common wrong workings: $\frac{100}{360} \times 480$, $\frac{100}{480} \times 100$, $\frac{100}{180} \times 480$

Common wrong responses: 133.3, 20.8, 266.7

Expected response: $\frac{100}{480} \times 360 = 75$

(b) *The sector angle for the people who liked tea is 60°.
Find the number of people who liked tea.*

This part was fairly done. Most candidates were able to find the number of people who liked tea, using the correct method. Some candidates even used ratio, using their answer from part (a), correctly. A few candidates found their answer by trial.

Common wrong workings: $\frac{60}{480} \times 360$, $\frac{60}{360} \times 100$, $\frac{480}{60}$

Common wrong responses: 45, 16.7, 8

Expected response: $\frac{60}{360} \times 480 = 80$

Paper 6880/03

General Comments

An analysis of candidates' performance suggests that the examination paper was generally challenging for most candidates. While no candidate achieved the maximum score of 80, the highest mark attained was 79. Relatively few candidates scored above 70, indicating that high performance was uncommon. A small number of candidates, approximately four, recorded a score of zero, and a few others obtained only single-digit marks. In addition, many candidates across the majority of examination centres scored below 40, further highlighting the overall difficulty of the paper.

The time allocated for the paper was enough since all the candidates seem to have used the allocated time efficiently.

Questions that proved to be difficult for the candidates were **Q3 (b), Q6, Q9, Q11, Q13 Q14, Q15, Q16 (b), Q18 (b), Q20 (b), 21, Q23 and Q24**. Most of these questions required more than one skill per question so most of the candidates failed to cope.

Questions that were easily accessible for candidates were **Q2, Q5, Q8, Q10, Q16 (a), Q17 (b), Q20 (a) and Q26**.

The candidates demonstrated a great improvement in their work presentation since they showed almost all the necessary working except in Q8 where they simply gave the correct answer. Candidates are advised to show all necessary working especially with questions which are allocated more than 1 mark.

Lack of skill on the use of the electronic calculator was still a problem. This was evident in **Q1, Q19 and Q25**.

The candidates easily lost marks in Q3 (b), Q6, Q12 and Q14 which required the skill of directed numbers.

There were few cases where candidates rounded the required answers to 2 significant figures instead of using 3 or more significant figures which was highly commendable.

Comments on specific questions**Question 1**

Candidates were required to evaluate $\sin^{-1}(\sqrt[3]{0.9})^2$.

This was poorly done as most candidates did not get the correct answer which is 68.8. Most of them interpreted this as $\sin^{-1}(0.9)$ which resulted to 64.2 or

$(\sin^{-1} \sqrt[3]{0.9})^2$ which gave them 5 610.56. This demonstrated lack of skill on the use of the electronic calculator. Some had truncated answers like 68.7 or 68.

Common wrong answers: 64.2, 5 610.56, 68.7 and 68

Expected response: 68.8

Question 2

The candidates were given the ratio of girls to boys as 8: 7 respectively and asked to express the number of girls as a percentage of the class.

This was generally well answered since most candidates were able to get the expected response of 53.3 even though some gave their answer as 53 (correct to 2 s.f) which was unacceptable. Very few had $\frac{8}{100} \times 15 = 1.2$

Common wrong answers: 53, 1.2

Expected response: 53.3

Question 3

Three vectors, $\underline{a} = \begin{pmatrix} -1 \\ -2 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$ and $\overrightarrow{CD} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ were given (drawn) on a grid.

(a) The candidates were required to draw $\overrightarrow{AB}$ on the same grid which is such that $\overrightarrow{AB} = 2\underline{a} + \underline{b}$ where point A was marked on the grid.

This part was not easily accessible to the candidates since some had $\overrightarrow{AB}$ drawn without an arrow or direction and some marked B without joining it to A and some showed the calculations of $2\underline{a} + \underline{b} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$ but failed to draw it on the grid. There was an exceptional case of those who had the correct vector but decided not to use the given A on the grid instead they marked their own A.

Common wrong answers: $\begin{pmatrix} 0 \\ -2 \end{pmatrix}$ on calculation, vector without direction, correct vector $\overrightarrow{AB}$ with own marked A.

Expected responses: correctly drawn $\overrightarrow{AB} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$

(b) In this part, the candidates were asked to find the values of h and k, given that $\overrightarrow{CD} = h\underline{a} + k\underline{b}$

This part proved to be the most difficult for the candidates. Most tried to formulate simultaneous equations but failed to come up with the correct ones or to solve the correct simultaneous equations but still obtained wrong values after making some arithmetic errors.

Common wrong answers: none

Expected response: $h = 1$ and $k = 2$

Question 4

The candidates were required to find the gradient and the y intercept of a line with equation: $y - mx = 5$, passing through (2, 9).

This question was fairly answered. Very few had $9 - 2x = 5$ after substituting (2,9) incorrectly into the equation instead of $9 - 2m = 5$ and hence they had $x = 2$ instead of $m = 2$. Some had the correct equation $9 - 2m = 5$ but obtained $m = -2$ (failed to solve it correctly). The y intercept was either given as (0, 5) or as an equation $y = 2x + 5$.

Common wrong answers: $m = -2$ or $x = 2$ and for the y intercept: (0, 5) or

$$y = 2x + 5$$

Expected response: gradient = 2

$$y - \text{intercept} = 5$$

Question 5

The question required the candidates to write $\overrightarrow{OB}$ as a column vector, given that

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ 7 \end{pmatrix} \text{ and } \overrightarrow{AB} = \begin{pmatrix} 5 \\ 3 \end{pmatrix}.$$

This was generally well answered since quite a number of them were able to add the two vectors to come up with the correct answer $\begin{pmatrix} 8 \\ 10 \end{pmatrix}$. However some failed to express their answers correctly and gave it as either

$$\frac{8}{10} \text{ or } \frac{8}{10} \text{ or } \left(\frac{8}{10}\right) \text{ which resulted to loss of marks. Very few had } \begin{pmatrix} 3 \\ 7 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \end{pmatrix}.$$

Common wrong answer: $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$ or $\frac{8}{10}$ or $\frac{8}{10}$ or $\left(\frac{8}{10}\right)$

Expected response: $\begin{pmatrix} 8 \\ 10 \end{pmatrix}$

Question 6

Candidates were given the function $h(x) + 14 = x + 36$ and expected to find $h^{-1}(2)$.

This question proved to be the most challenging to the candidates. They could not correctly make $h(x)$ the subject of the formula. Some had the correct $h(x)$ and the correct equation for the inverse; $x = y - 22$ but could not change it to the correct inverse function notation and got confused afterwards hence they lost marks. Some had the correct inverse, $h^{-1}(2) = 2 - 22$ but gave their answer as 20 instead of -20 (failed to use their calculator or handle directed numbers). Some confused the notations of $h(2)$ and $h^{-1}(2)$ thus they had $h^{-1}(2) = 2 + 22 = 24$

Common wrong answer: 20 and 24

Expected response: -20

Question 7

A bar chart representing the shoe sizes (1, 2, 3, 4 and 6) of 25 learners was given showing the following frequencies.

Shoe size	1	2	3	4	5	6
Frequency	3	2	5	6		6

(a) The candidates were asked to complete the bar chart to show the bar for size 5.

This part was generally well answered by almost all the candidates. Very few had a wrong bar drawn without even showing the calculation of how it was obtained.

Common wrong answer: none

Expected response: correct bar at a height of 3

(b) *The candidates were expected to find the median size.*

This part was not well answered. Some had $\frac{3+4}{2} = 3.5$ or $\frac{1+2+3+4+5+6}{6} = 3.5$

Common wrong answer: 3.5

Expected response: 4

Question 8

The candidates were given the mean of the marks for a History class as 37 and the total sum of their marks as 1554 and required to calculate the number of learners in the class.

This question was easily accessible for most of the candidates. The correct answer, 42 was common although some did not even show the necessary working. Very few only left it as: $\frac{1554}{x} = 37$ and did not solve for x .

Common wrong answer: none

Expected response: 42

Question 9

The candidates were given three sets of numbers.

$\mathcal{E} = \{\text{natural numbers}\}$

$Q = \{2, 4, 6, 8 \dots\}$

$R = \{x: x = t^2 \text{ where } t \text{ is a positive integer}\}$

(a) *The candidates were asked to name the type of numbers in*

(i) set R

(ii) set Q

The whole question proved to be the most inaccessible for most of the candidates. The use of R and Q in naming the sets caused a confusion since Q is mathematically used to denote rational numbers and R for real numbers. Other letters could have been used instead to avoid this confusion. Candidates failed to answer this part questions correctly since R was misinterpreted for real numbers and Q for irrational numbers. Only a few recognized $R = \{x: x = t^2 \text{ where } t \text{ is a positive integer}\}$ as square numbers.

Common wrong answer:

(i) Real numbers

(ii) Irrational numbers

Expected response:

(i) Square numbers

(ii) Odd numbers

(b) The candidates were asked to list the lowest 3 elements of $R \cap Q'$.

This part was not well done by the majority of the candidates. Those candidates who were able to get (a) correctly gave their answers as 9, 25, 49 instead of 1, 9, 25 which showed that candidates did not appreciate 1 as a square number.

Common wrong answers: 9, 25, 49

Expected responses: 1, 9, 25

Question 10

The candidates were expected to find the total number of men and women in a meeting with 25 men given the ratio of men to women as 5: 4 respectively.

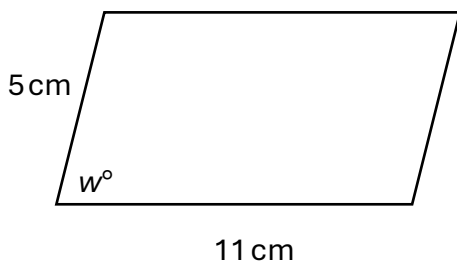
This question was inaccessible to most candidates yet it was easy. Majority of the candidates failed to understand the question. Some simply gave their answer as 20(number of women only) yet others gave 11 women ($\frac{4}{9} \times 25$) and 14 men ($\frac{5}{9} \times 25$).

Common wrong answer: 20 or 11 women & 14 men

Expected response: 45

Question 11

A parallelogram of area 44 cm^2 , base of 11 cm long and slant side 5 cm long was given.



NOT TO

The candidates were asked to find the value of w .

This was inaccessible to most candidates. They simply treated the figure as a triangle and used the trigonometric formula for calculating area of a triangle as follows:

$44 = \frac{1}{2}(5)(11)\sin w$ and failed to get the answer. Some used $44 = \frac{1}{2}(11 + 5)\sin w$. Very few used the correct formulas $22 = \frac{1}{2}(5)(11)\sin w$ or $\sin w = \frac{4}{5} = 53.1$

Common wrong answers: 53

Expected response: 53.1

Question 12

Candidates were given $\begin{pmatrix} a & -1 \\ 4 & 2 \\ -4 & 1 \end{pmatrix} \begin{pmatrix} -2 \\ 5 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ b \end{pmatrix}$ and expected to find the values of a and b .

This question was generally well done. Most candidates were able to get the correct answers: $a = -1$ and $b = 13$ (most commonly seen). Some had problems in handling directed numbers when solving for the value of

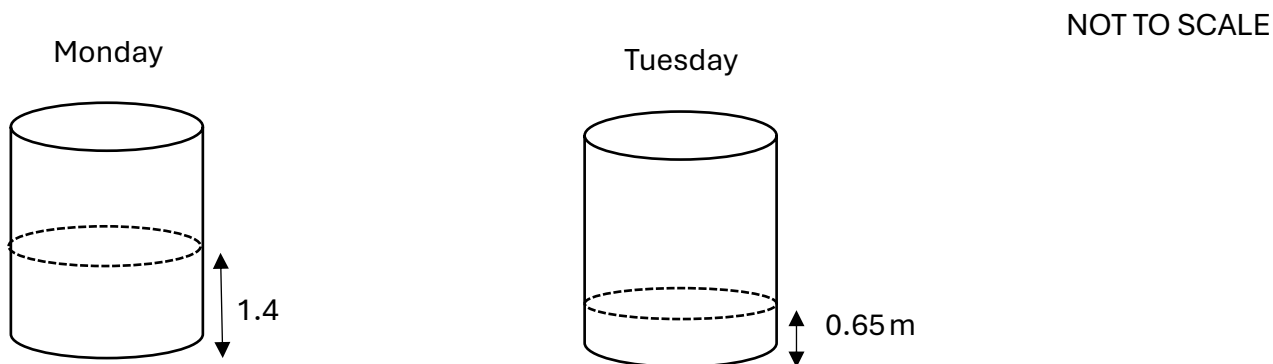
a in the equation, $-2a - 5 = -3$ which led to a wrong answer of 4. Some candidates were seen finding the value of a by inspection without showing the equation, $-2a - 5 = -3$ which was required.

Common wrong answer: 4 or 1

Expected response: $a = -1$ and $b = 13$

Question 13

Candidates were given two cylindrical tanks with water of depth 1.4m on Monday and a depth of 0.65 m on Tuesday. The volume of the water in the cylindrical tank on Monday was 3.56m^3 .



This question was not well done since most candidates did not realize that the cylinders were not similar to each other but congruent. So of the candidates converted the ratio of sides to ratio of volumes and used that ratio to calculate the volume of the water on Tuesday as follows:

$$\frac{3.56(0.65^3)}{1.4^3} = 0.356$$

Some of the candidates used the wrong formula for volume of cylinder e.g. $V = \pi r h$ instead of $V = \pi r^2 h$ to calculate r which gave them $r = 0.809$ instead of 0.89968 .

Common wrong answers: 0.356

Expected response: 1.65

Question 14

The points $A(-1, 2)$, $B(3, a)$ and $C(5, 8)$ lying in a straight line were given. The candidates were asked to find the value of a .

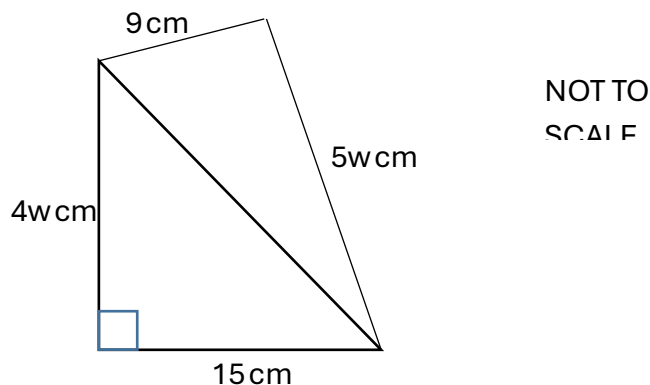
This proved difficult for the candidates since the concept of collinear points did not seem to be remembered. Those who remembered it, calculated the gradients and failed to use them whereas some had difficulties dealing with directed numbers e.g. $5 - (-1)$. Some tried to solve graphically but failed since their grid was not up to scale. Some calculated midpoints and magnitudes which clearly showed that they were totally lost.

Common wrong answer: 5 from wrong graph

Expected response: 6

Question 15

The candidates were expected to find the value of w on the given diagram with two right angled triangles.



This question was poorly done by most of the candidates. They either left it blank because they failed to recognize that the triangles had a common hypotenuse or applied Pythagoras wrongly: $4w^2 + 15^2 = 9^2 + 5w^2$ which led to $w = 12$ or $\sqrt{9^2 + 25w^2} = \sqrt{15^2 + 16w^2}$ which simplified to

$9 + 5w = 15 + 4w$, hence $w = 6$ or $\sqrt{9^2 + 25w^2} = \sqrt{15^2 + 16w^2}$ which gave them $9 + 25w = 15 + 16w$, $9w = 6$ hence $w = \frac{2}{3}$

Common wrong answer: 12 or 6 or $\frac{2}{3}$

Expected response: 4

Question 16

(a) $p = \frac{5x}{3}$ and $q = \frac{7x}{15}$ were given and the candidates were required to write $2p + q$ in terms of x .

This part was very poorly done. Most of the candidates would substitute correctly and have $2\left(\frac{5x}{3}\right) + \frac{7x}{15}$ then

simplify wrongly to $\frac{10x}{6} + \frac{7x}{15} = \frac{19x}{15}$ and some treated $2p$ as a mixed number i.e. $2\frac{5x}{3} + \frac{7x}{15}$ which simplified to

$$\frac{11x}{3} + \frac{7x}{15} = \frac{62x}{15}$$

Common wrong answers: $\frac{19x}{15}$ or $\frac{62x}{15}$

Expected Response: $\frac{57x}{15}$

(b) The candidates were expected to write m in terms of x , given that $y - x = 3$ and $2m + y = 5$.

This part was most inaccessible to most candidates since they failed to make y the subject of the formula in one of the equations properly then substitute on the other one i.e. $y = 3 - x$ instead of $y = 3 + x$, which gave them $2m + 3 - x = 5$ and yielded $\frac{2+x}{2}$ and some would then make a lot of arithmetic errors in the process and end up with answers like $\frac{x-2}{2}$.

Common wrong answers: $m = \frac{5-3-x}{2}$ or $\frac{2+x}{2}$ or $\frac{x-2}{2}$

Expected Response: $\frac{2-x}{2}$

Question 17

The question required the candidates to solve the equations.

(a) $x + 5 = -\frac{6}{x}$

This part was not well answered. Some failed to multiply all the terms by x and ended up having $x^2 + 5 = -6$ instead of $x^2 + 5x = -6$. Those who had the correct quadratic equation:

$x^2 + 5x = -6$ failed to factorise correctly to $(x + 3)(x + 2) = 0$ but instead had $(x + 3)(x - 2) = 0$ or $(x+6)(x+1) = 0$. Those who attempted to use the quadratic formula for solving did not use it correctly since their division

line was short i.e. $-5 \pm \frac{\sqrt{1}}{2}$ instead of $\frac{-5 \pm \sqrt{1}}{2}$. Those who solved by completing the square got lost along the way and lost all the marks.

Common wrong answer: -1 or -6, -3 or 2 or vice versa

Expected responses: $x = -2$ or $x = -3$

(b) $\frac{1}{3} = \frac{3x-4}{6}$

This part was generally well done by the majority of the candidates. However those who cleared their fractions by 18 to obtain $3(3x - 4) = 6$ could not properly open the brackets to obtain

$9x - 12 = 6$ but instead got $9x - 4 = 6$.

Expected response: $x = 2$

Question 18

The candidates were expected to factorise the expressions.

(a) $5k^2 - 35k$

This part was well answered. However some factorised it partially and obtained $5(k^2 - 7)$ or $k(5k - 35)$ yet some attempted to factorise it as a difference of two squares and they had $5(k+7)(k - 7)$.

Common errors: $5(k^2 - 7)$ or $k(5k - 35)$ or $5(k+7)(k - 7)$

Expected response: $5k(k - 7)$

(b) $3m^2 - 13m + 12$

This part was fairly done although a few of the candidates tried to *solve the expression using the quadratic formula instead of factorizing it. Very few interchanged the variable m with x and eventually had*

$(x - 3)(3x - 4)$ instead of $(m - 3)(3m - 4)$.

Common wrong answers: $(x - 3)(3x - 4)$, $m = 3$ or $m = \frac{4}{3}$

Expected response: $(m - 3)(3m - 4)$

Question 19

The candidates were asked to state the ratio of old to new value of a car whose value has decreased by 15% over a period of time.

Well answered by most candidates. Only a few failed to use their calculator to subtract 15 from 100 to get 85 but obtained 75 instead which led to easy loss of marks.

Common wrong answers: 85 or 0.75 or 75%

Expected response: $0.85 \left(\frac{17}{20} \right)$

Question 20

The candidates were given a table with 3 sequences.

1	2	3	4	5	...	n
9	16	25	36	a	...	l
8	15	24	b	48	...	m

(a) *The candidates were asked to find the values of a and b .*

Most of the candidates were able to get the correct values of a and b as 49 and 35 respectively.

Common wrong answers: none

Expected response: $a = 49$ and $b = 35$

(b) The candidates were required to write l and m in terms of n .

This part was not easily accessible to most of the candidates. The correct expressions for L and m were rarely seen. They were left in numerical form: $l = 81$ (9^2) and $m = 81 - 1 = 80$. Different expressions for l in the form an^2 were also seen.

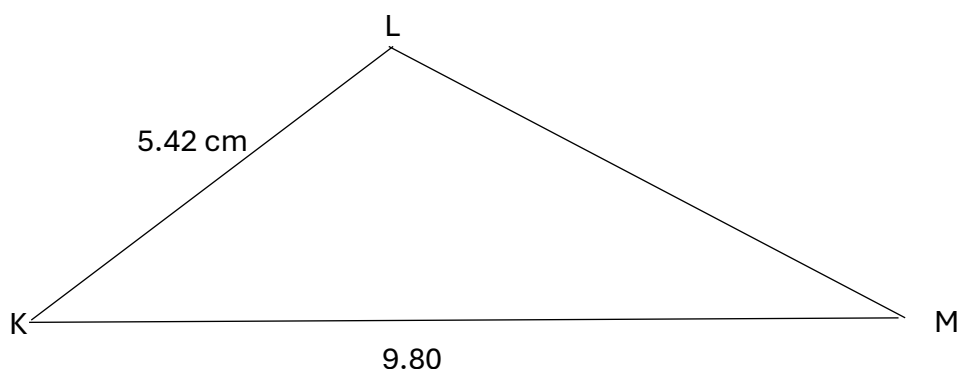
Common wrong answers: $l = 81$, $m = 80$

Expected response: $l = (n + 2)^2$ or $n^2 + 4n + 4$

$$m = (n + 2)^2 - 1 \text{ or } n^2 + 4n + 3$$

Question 21

Candidates were expected to calculate the size of angle KLM on the given triangle with a perimeter of 21.64 cm.



This question was not well performed. Most candidates were able to find side LM as 6.42 except for a few who made some arithmetic errors. Very few were able to use the cosine formula correctly to calculate the required angle but failed to simplify and ended up having $96.04 = 1 \cos L$ yet some had $(+) 2mk \cos L$ and some used $(-) 2mk \sin L$. Quite a number of them used the cosine formula to calculate a wrong angle K:

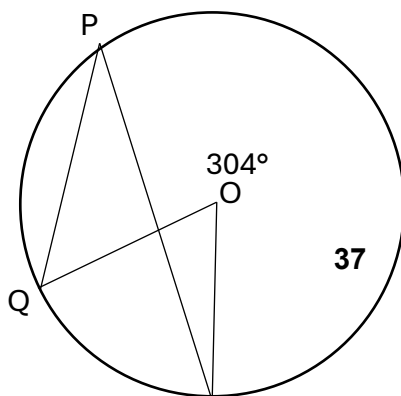
i.e. $6.422 = 5.422 + 9.82 - 2(5.42)(9.8) \cos K$

Common wrong answers: 37.6

Expected response: 111.4°

Question 22

The question required the candidates to calculate angle QPS on a circle with centre O and reflex angle $QOS = 304^\circ$.



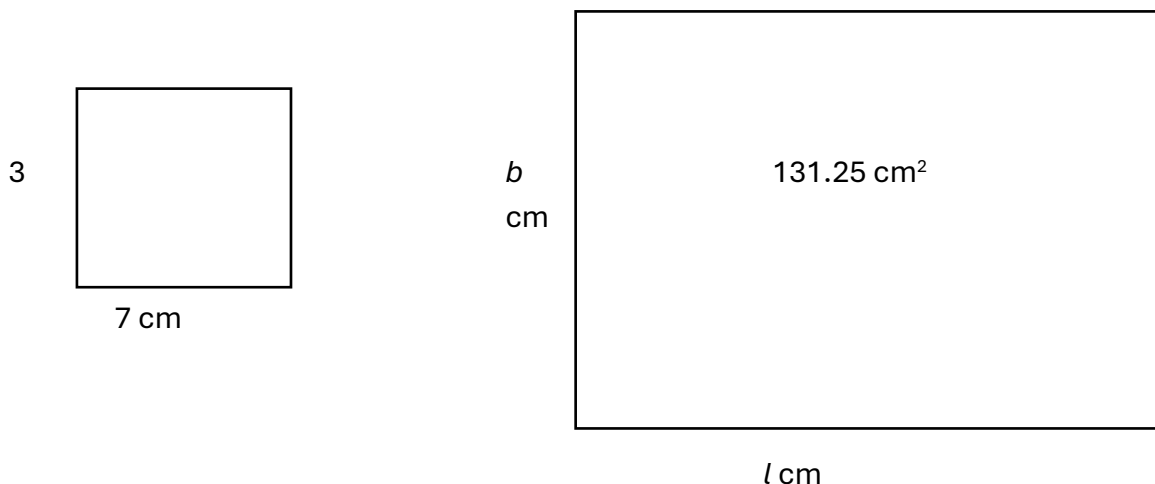
This question was generally well done. Some treated angle QOS and angle QPS as angles of the same segment therefore equal to 56°

Common wrong answers: 56°

Expected response: 28°

Question 23

The candidates were given two similar rectangles. They were also given the area of the larger rectangles as 131.25 cm^2 .



The candidates were asked to work out b and l .

This question was the most poorly done. The majority of them failed to convert the ratio of areas to ratio of sides but simply used the ratio of areas as it is to calculate b and l which led to $l = 43.75$ and $b = 18.75$

Common wrong answers: $b = 18.75$ and $l = 43.75$

Expected response: $b = 7.5$

$$l = 17.5$$

Question 24

The question required the candidates to find the probability that exactly two of the here boys buy the paper on a particular day under the given conditions;

The probability that Musa buys the newspaper on any day is $\frac{3}{8}$.

The probability that Themba buys the newspaper on any day is $\frac{3}{4}$.

The probability that Mandla buys the newspaper on any day is $\frac{1}{2}$.

This question was the most poorly done. The question was either left blank or had wrong tree diagrams. Those who had correct tree diagrams failed to choose the correct outcomes to add. Some demonstrated the weakness of not understanding exactly two by simply writing $\frac{3}{8} \times \frac{3}{4} \times \frac{1}{2} = \frac{9}{64}$ or $\frac{3}{8} \times \frac{3}{8} = \frac{9}{64}$ or

$$\frac{3}{8} \times \frac{3}{4} + \frac{3}{4} \times \frac{1}{2} + \frac{3}{8} \times \frac{1}{2} = \frac{27}{32}$$

Common wrong answers: $\frac{27}{32}$ or $\frac{9}{64}$

Expected response: $\frac{27}{64}$

Question 25

The candidates were given a table representing the amount of money a group of 30 people spent on airtime in a week.

Money(E)	$0 < x \leq 30$	$30 < x \leq 50$	$50 < x \leq 60$	$60 < x \leq 70$	$70 < x \leq 75$
Frequency	7	10	8	3	2

The candidates were asked to calculate an estimate of the mean.

The majority of the candidates were able to answer this question. Some had

$$\frac{30(7) + 20(10) + 10(8) + 10(3) + 5(2)}{30} = 17.7 \text{ (using differences between upper and lower class) yet some had}$$

$$\frac{15(7) + 40(10) + 55(8) + 65(3) + 73(2)}{30} = 42.9 \text{ (rounded last midvalue 72.5 to 73) yet some used the upper}$$

$$\text{class boundaries: } \frac{30(7) + 50(10) + 60(8) + 70(3) + 75(2)}{30} = 51.7.$$

Some knew that they had to use midvalues but failed to use the calculator correctly: instead of $\frac{30 + 50}{2} = 40$

for midvalue of $30 < x \leq 50$, they had $30 + \frac{50}{2} = 55$ (on their calculator) and their mean was calculated as

$$\frac{15(7) + 55(10 + 80(8) + 95(3) + 107.5(2))}{30} = 59.8$$

Common wrong answers: 42.9 or 17.7 or 51.7 or 59.8

Expected response: 42.83

Question 26

The candidates were asked to factorise the expression $5ab - 5ac + 4b - 4c$

This question was well done by the majority of the candidates. The only challenge with some was omitting the sign during grouping factorization as follows: $5a(b - c) 4(b - c)$ which led to

$(5a - 4)(b - c)$ instead of $(5a + 4)(b - c)$. Some led their answers partially factorized as either $5(ab - ac) + 4(b - c)$ or $5a(b - c) + 4(b - c)$ or $5a(b - c) + 2(2b - 2c)$

Common wrong answers: $(5a - 4)(b - c)$ or $5(ab - ac) + 4(b - c)$ or $5a(b - c) + 4(b - c)$ or

$5a(b - c) + 2(2b - 2c)$

Expected response $(b - c)(5a + 4)$

Paper 6880/04**General comments**

A total of 8,058 candidates sat for the 2025 Mathematics Paper 4 examination. Overall, performance was affected by how answers were presented rather than a complete lack of understanding. Many candidates were able to carry out the correct calculations, but a large number gave their final answers to only two significant figures instead of the required three, which led to a loss of marks.

There were also clear signs that some candidates struggled to interpret questions carefully. For instance, in Question 1(a), key wording such as “did not choose pork” was often overlooked, with many candidates answering as if the question had asked for those who “chose pork.”

In algebra, understanding was somewhat inconsistent. Some candidates confused basic processes, such as collecting like terms, with transposing terms across an equation. In addition, questions that required several steps proved challenging. While many candidates started off correctly and showed appropriate working, they often erased these steps and left only the final answer, which made it difficult to award full marks.

Two candidates obtained the 100% score of 120 in the paper. About 6% of the candidates managed to score at least 100 marks in the paper. Only 20% of the scores were below the 20 marks. There were less than 20 candidates who scored no points in the examination. The group struggled with questions that required the use of a combination of methods and multiple steps. This was evident in questions such as question 12, 13, 11, 9 and 10. Analytical skills that enable candidates to effectively identify the full requirements of questions seem to still need further development in candidates. In question 8a, the density was given with the relevant units, which could be used as a clue toward the formula needed, yet some candidates came with unfamiliar equations for finding density. There was a great improvement in demonstrating content mastery by some candidates, hence the scoring of all marks in a question.

This paper requires candidates to be skilled with the algebra rules applicable in any type of formulae such as those used in trigonometry, rates of change and area and volume of solids. They needed to be familiar with correctly interpreting question requirements and reflect on the reasons for each choice of method they use. It is recommended that candidates master data meaning and analysis in statistics and probability

The questions that proved to be optimally challenging to candidates were questions 6(a)(i), (ii)(using matrices for combined transformations), 8a,b(using capacity and rate formulae such as density and speed to calculate suggested quantities), 10a,b(making decisions on methods to find length and angles required for calculating the area of a non-right angled triangle), 12a,b (finding the radius from the volume of a sphere and volume of similar figures), 13a(ii), (finding the x-coordinate of the turning points of cubic functions), and question 14(a)(b)(conditional probability of combined events) The only questions that candidates found favourably accessible were question 3a(ii)(solving a linear inequality), 3c(forming linear simultaneous equations from words) and 9a,b,c(expanding a single bracket and division by an algebraic fraction).

The ambidexterity shown through the various correct approaches used by candidates in answering questions was a marvellous improvement in content mastery. Educators are encouraged to train candidates on the skill of showing all working used to build numerical values essential to the final solution of a question. Training candidates on the dos and don'ts of solving equations is required.

Comments of specific questions

Question 1

- a) The question required candidates to find the percentage of delegates that did not choose pork. The most common error resulted from calculating the percentage of those who chose pork.

Common incorrect response was 26.7%

The expected correct response was $\frac{(3 \times 48) + 120}{360} = 73.3\%$

- b) (i) The question was moderately well answered. Instead of finding a fraction of the total number of people, candidates found the number of people only. The common wrong answer was 20.

The expected correct answer was $\frac{20}{60}$.

(ii) Candidates were able to find the correct number of people who liked Amapiyano music. They however, forgot to express that as a fraction of the people in the whole group. The frequently occurring wrong responses were; 15.

The expected correct response was $\frac{15}{60}$.

Question 2

- (a) A majority of candidates answered the question well. The common incorrect error was placing correct values in the wrong positions of the Venn diagram. **The anticipated correct answer was: the values, 5, 11 and 8 placed in the correct regions .**

- (b) (i)

(ii) The wrong response of 29, was a result of including the members who could speak all three languages. **The expected answer** was for members that could speak **exactly two** of the three languages, **which was: 22**

- (c) (i) One of the least well performed question was this one. The candidates gave a solution for tourists who could speak **Portuguese only**. The common wrong answer was 42.

The expected answer was for the total number of tourists who could speak Portuguese

The expected final answer was: 60.

- (d) (ii) This was one of the well performed question. The few errors were from those candidates who calculated the number of elements in set E. The resulting wrong response was: 53.

The expected correct response was 29.

Question 3

- (a) (i) Candidates could hardly arrive at the correct solution to this question. They either omitted essential steps in the working, or use wrong methods of solving equations, such as dividing by the variable they were solving for, instead of factorising. One wrong solution was: $5x^2 - 2x = 0$

$$\frac{5x^2}{x} = \frac{2x}{x}$$

$$5x = 2$$

$$x = \frac{2}{5}$$

The expected correct solution was:

$$5x^2 - 2x = 0$$

$$x(5x - 2) = 0$$

$$x = 0 \text{ or } x = \frac{2}{5}$$

(ii) A very accessible question by most candidates was this one. The only error was replacing the inequality sign with an equal sign resulting in a **common wrong answer of $x=10$** .

The expected correct response was $x < 10$.

(b) The mistake made by some candidates was, forgetting to multiply $(r - w)by - 1$ in $-(r - w)$. The resultant incorrect response was $r = \frac{35-w}{16}$.

The expected correct answer was: $r = \frac{35+w}{16}$

(c) Finding the expected simultaneous equations described in the words proved easy for the candidates. Minimal mistakes were noted in the solution steps of solving the equations.

The expected response was; $2x + 3y = 54$ and $2.5x + 3.5y = 65$ yielding the final solution of; $x = 12$ and $y = 10$.

Question 4

(a) Most of the candidates found this question very difficult. They substitutes $g(3)$ into $f(t)$ instead of $f^{-1}(t)$. The incorrect solution resulting was $(5 \times 9) - 19 = 26$.

The anticipated response was: $\frac{28}{5} = 5.6$

(c) Candidates demonstrated knowledge of evaluation of functions at a given point. The frequently occurring incorrect response was a result of inability to effectively substitute the 2-term expression in the place of a single variable. They substituted $g(t)$ into t , instead of substituting $\frac{1}{2}t - 6$ into $g(t)$. The common incorrect answer was: $\frac{1}{2}(2t + 3) - 6 = t - 4.5$.

The expected correct answer was: $\frac{1}{2}(2t - 6) + 3 = t - 9$

(d) The question expected candidates to identify the pattern used in (b) to solve the equation. The few errors in responses arising only from taking only the positive value of $\sqrt{9}=3$

The expected correct answer was: $\pm\sqrt{9} = 3 \text{ or } -3$

Question 5

(a) This question was well answered. Errors were noticed in wrongly labelled axes that resulted into 20 outcomes instead of 25.

The expected correct answer was: a correctly labelled possibility space diagram with 25 outcomes

(b) (i) Although some possibility space diagrams were wrong, the candidates demonstrated good interpretation skills. The common incorrect response was $\frac{2}{20}$.

The expected correct answer was: $\frac{2}{25}$

- (ii) The question was correctly answered by a majority of candidates.

The common wrong answer was: $\frac{3}{20}$.

The expected correct response was: $\frac{3}{25}$.

Question 6

- (a) (i) A correct method of using matrices for transformation was shown by most candidates. The points were lost due to error in multiplication and addition of directed numbers. The most frequently occurring error was, a triangle rotated 90° clockwise but using a different centre from (0.0).

The expected correct response was: *A triangle with correct vertices*

(ii) Most candidates found this question very difficult. Finding a matrix for combined transformation was not easy. Those who tried step by step application of the transformation involved were able to correctly obtain the first image, but could not find the expected enlargement. The error was a result of inability to find the correct matrices of the described transformations. Some candidates just performed an enlargement of the given triangle without effecting the first transformation.

The expected correct response was: a triangle that was first reflected over the y-axis, then enlarged with scale factor -2

- (b) Candidates used wrong name for a translation. They presented the vector as a fraction and interchanged the direction of the horizontal and vertical movements of the translation vector. Therefore, the common incorrect response was: translocation by vector $\begin{pmatrix} 1 \\ -4 \end{pmatrix}$

The expected correct response was: translation by vector $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$

Question 7

- (a) The difficulty in most candidates was translating the variation statement into a correct mathematical equation. The square of V was written as the square root of V by some candidates.

Frequent wrong responses include: $T = \frac{k}{\sqrt{v}}$ and $\frac{125}{v^2}$.

The expected correct answer was: $T = \frac{1000}{v^2}$.

- (b) This was one question which was moderately answered correctly. Wrong responses arose from division by V instead of the square of V . Some correct working would be wrong at the last step, where they needed to find the square root of the v^2 expression.

One wrong solution was: $a = \frac{1000}{4} = 250$ and $b = 125 \div 40 = 3.125$.

The expected correct answer was: $a = \frac{1000}{16} = 62.5$ and $b = \sqrt{\frac{1000}{40}} = 5$.

Question 8

- (a) The question required candidates to simply find the mass of a liquid and use the density formula to calculate its volume. Candidates had who lost points were those who forgot to first find the mass of the liquid.

The common wrong response was: $v = \frac{400}{3.05} = 131$.

The expected correct response was: $v = \frac{125}{3.05} = 41.0$.

- (b) (i) Candidates hardly paid attention to the detail on finding an expression for **time**, they just substituted given values into the formula for speed.

The resulting wrong response was: $72 = \frac{x}{t}$

The expected correct answer was: $t = \frac{x}{72}$

- (ii) Another hardly accessible question was this one. A large number of candidates were able to find the expression for distance, but they could not make t the subject of their equation.

The common incorrect answer was: $\frac{10+x}{54}$ and $10 - x$.

The correct response was: $\frac{10-x}{54}$

- (iii) This question proved to be very inaccessible for a majority of the candidates. They could not identify equivalent expressions for time. Some added the expressions instead of equating and rearranging. The resulting wrong responses were abandoned expressions that did not end with numerical values of x and t .

The anticipated corrected response was: $\frac{x}{72} = \frac{10-x}{54}$ leading to $x = 5.71$ and $t = 0.0794$.

Question 9

- (a) Candidates were required to remove the brackets and collect like terms. They correctly removed the brackets to get; $35 - 5x + 3x + 12$. The error they committed was subtracting 12 when collecting like terms instead of adding as if they were dealing with an equation.

The resulting common error was: $35 - 12 - 2x$ yielding, $23 - 2x$

The expected response was: $47 - 2x$.

- (b) Errors in collection of like terms and inability to identify common factors after trying to factorise expressions were the reason most candidates lost points.

The expected correct response was: $\frac{(x+6)(x-1)}{2(x+6)}$ resulting to; $\frac{x-1}{2}$.

- (c) A large number of candidates could not find a correct solution to the problem due to inability to interpret division by a fraction. A variety of wrong responses were given.

The expected correct answer was: $\frac{3-3x}{2}$.

Question 10

- (a) **The inaccessibility** of the question to most candidates was a result of trying to calculate the area using the sine formula before finding the lengths and angle applicable to the formula.

The frequently occurring incorrect response was: $\frac{1}{2} \times 9 \times 12.3 \sin 105^\circ = 53.5$.

The expected correct response was: to first find angle C, then the sine formula for area: $\frac{1}{2} \times 9 \times 12.3 \sin 30^\circ = 27.675$.

- (b) Although a lot of candidates could not earn the points in this question, those who were able demonstrated very impressive approaches of arriving at the correct answer. Some used the area in (a) to find the length of AB; ($\frac{1}{2} \times 9 \times AB \times \sin 105^\circ = 27.675$). One of the errors was using sine in a cosine formula context, $c^2 = a^2 + b^2 - 2ab \sin c$.

The expected correct answer was: 6.37.

Question 11.

- (a) The correct solution depended on a correct radius. Candidates used the formula for the area of a circle instead of the circumference of a circle. The common error was therefore: $6\pi = \pi r^2, r = 2.45$. This radius produced a wrong volume of 23.1

The expected correct response was: $2\pi r = 6\pi, r = 3$ which yields $Volume = 34.87$.

- (b) The question was scarcely well answered. The common error committed by most candidates was trying to force the use of an unknown-curved surface area of a cone ($\pi r l$) to find l . **The expected response was:** $l = 4.76$

Question 12

- (a) Almost 85% of responses were incorrect. When calculating the area, candidates used the correct volume formula to find the radius. However, they subtracted a number instead of dividing by it.

The common error was; $\frac{4}{3}\pi r^3 = 904.8, r = \sqrt[3]{904.8 - 4.1887} = 9.65$ The mistake revealed a slight decline in the skill of changing the subject of a formula.

The expected correct response was; $r = 6$ and **surface area = 452.4**

- (b) The question was hardly correctly answered. Candidates were expected to find the ratio of volumes of the 2 similar solids, then calculate the required volume. However, most candidates used the ratio of the heights of the solid 2:3.

The most common wrong answer was, $volume = \frac{424 \times 3}{2} = 636$.

The expected correct response was to use 8:27 to find, $volume = \frac{424 \times 27}{8} = 1431$

Question 13

- (a) (i) A large number of candidates found this question slightly inaccessible. Candidates were able to find the derivative of the cubic function, but committed errors in manipulating the gradient through substitution of the x-coordinate. Various wrong answers were commonly given as the equation of the tangent.

The expected correct response was: $\frac{dy}{dx} = 3x^2 - 4x - 9$, *leading to the equation; $y = -5x - 8$.*

- (ii) The error made was treating the function as quadratic yet it was cubic.

The most frequent wrong response was using: $x = \frac{-b}{2a}$. Those who managed to use the correct method, lost marks from truncating their answers to $x = 2.4$ or $x = 1.2$

The expected correct response was; $x = \frac{4 \pm \sqrt{124}}{6}$ *yielding $x = 2.52$ or -1.189 .*

- (b) A majority of candidates successfully found the second derivative function. They however, could not correctly classify the turning points.

The popular wrong responses were; $y'' = 6x$, *A is a minimum and B was a maximum.*

The expected correct response was: $y'' = -6x$, *A is a maximum and B was a minimum.*

Question 14

- (a) One of the most inaccessible question to this candidature was this one. The correctly found the probabilities for finding each of the letters in TEA, but when performing the combined conditional probability, they used the same denominator.

The common wrong answer was: $\frac{3}{9} \times \frac{1}{9} \times \frac{2}{9} = \frac{6}{729}$.

The expected correct answer was: $\frac{3}{9} \times \frac{1}{8} \times \frac{2}{7} = \frac{6}{504}$.

- (b) The question proved to be a challenge to candidates. The less than 3% candidates who were able to get the question correct illustrated several unique and good approaches. Some listed all the outcomes and carefully obtained correct results. Those who used elimination and permutation were also very good.

The expected correct answer was: $\frac{6}{9} \times \frac{5}{8} \times \frac{4}{7} = \frac{120}{504} = \frac{5}{21}$

Question 15

- (a) Almost all the candidates were able to get the cumulative frequencies required.

The expected response was: 5 and 80.

- (b) This question was moderately well answered. The few errors include interchanged axes, drawing polygons, missing the last point and leaving one point not joined.

The anticipated correct response was a curve passing through all correctly plotted points.

- (c) (i) The estimate of the median was well done by the group of candidates.

The expected correct response was; 56 ± 2

- (ii) A good number of candidates answered this question correctly. The mistake made was, subtracting the lower quartile position from the upper quartile position. They would then find an interquartile range-position then their interquartile.

This wrong method used was: $IQR = U.Q \text{ position} - L.Q \text{ position} = 75 - 25 = 50$. After this they located the corresponding mark of 56 as a common wrong answer.

The expected solution was: $IQR = U.Q - L.Q = 68 - 43 = 25 \pm 2$.

